

State Update

We define the state update operation $[v \mapsto n]$ on states as follows

$$\begin{aligned} s[v \mapsto n](u) &= n && \text{if } u = v \\ &= s(u) && \text{otherwise} \end{aligned}$$

where v and u are variables, and n is a number.

For example, with $s_1 = (x \mapsto 4, y \mapsto 5, z \mapsto 6)$

we have that $s_1[y \mapsto 88] = (x \mapsto 4, y \mapsto 88, z \mapsto 6)$